Namrata Paikekar

Data Scientist

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Pune, India

SUMMARY

I am an experienced Data Scientist specializing in data mining, statistical modeling, machine learning, NLP, deep learning, time series forecasting and generative AI. Proficient in Python, SQL and visualization libraries like seaborn, plotly, matplotlib, with a strong track record of delivering insights that drive business impact.

SKILLS

Python | ML Algorithms | Artificial Intelligence | Clustering | Data Mining | Data Science | NLTK | Machine Learning | Deep Learning| Linear/Logistic Regression | Random Forest | SVM | Decision Trees | NLP | Spacy | Naive Bayes | EDA | Data Visualization | Flask | Keras | K-means | Matplotlib | MSSQL | Numpy | Pandas | Plotly | Pytorch | VSCode | Scikit-learn | Scipy | Gen AI | LLMs | Langchain | Hugging Face | transformer models | AWS | Git

EXPERIENCE

MD India Health Insurance TPA Pvt. Ltd.

Oct 2024 – Present Pune, India

Data Scientist

- Understand problem statements, collect relevant data, perform exploratory data analysis (EDA), develop predictive models, fine-tune algorithms, evaluate model performance, and deploy solutions.
- Collaborate closely with the development team for seamless API integration of machine learning models into production systems.
- Implement classical machine learning and deep learning algorithms to address complex business challenges effectively.
- Currently working on cutting-edge technology Generative AI projects, focusing on Retrieval-Augmented Generation (RAG) systems.

Neosoft Systems and Cloud Services

Apr 2021 – Oct 2023 Pune, India

Junior Data Scientist

- Built and optimized supervised ML classification models (Logistic Regression, SVM, Random Forest, XGBoost) in Python/SQL environments to support predictive use cases.
- Performed comprehensive data cleaning, EDA, and feature engineering to enhance model performance and reliability.
- Engineered CNN-based deep learning models for computer vision tasks, advancing image analysis capabilities in production settings.
- Collaborated with cross-functional teams to translate business requirements into ML solutions, define success metrics, and iterate on model performance through hyper-parameter tuning and feedback loops.

Anand Impex Pune.

Jan 2019 – Mar 2021 Pune, India

PROJECTS

Tariff Digitization
Technologies: Python, OCR libraries (Unstructured, PyTesseract, PaddleOCR, Tabula), MSSQL
Project Description:
Developed a tariff digitization solution to extract and digitize the scanned SOC documents and store extracted data into structured and standardized format into MSSQL database for easy access and analysis. Leveraged different OCR libraries and text-processing techniques to convert non-searchable PDFs into structured SQL tables. Standardized hospital room categories by mapping them to unified formats, enabling accurate and efficient processing.

AI-Enabled Intelligent Coding Assistant for medical professors
Technologies: RAG, Python, MSSQL, Pandas, FAISS, BioGPT (Hugging Face Transformers), Pickle
Project Description:
Developed an AI-powered Retrieval-Augmented Generation (RAG) pipeline for healthcare domain data. The project integrated ICD–PCS codes and procedure descriptions stored in MSSQL with semantic search and generative AI. This solution enabled faster and more accurate medical coding assistance, treatment understanding, and decision support for healthcare professionals.

Advanced Fraud detection System for insurance claims processing
Technologies – Python, SQL, Scikit-learn, XGBoost, Random Forest, Logistic Regression, Pandas, NumPy, Matplotlib, Seaborn
Project Description:
Developed a hybrid fraud detection framework for insurance claims by combining rule-based anomaly triggers and supervised machine learning. Claims were first passed to catch the behavioural patterns using medical and operational triggers. Filtered high-risk claims were then passed into supervised ML models to minimize false negatives and ensure higher fraud capture rates. The system significantly improved fraud detection accuracy while maintaining business interpretability.

Key Contributions:

- Built end-to-end AI/ML solutions spanning tariff digitization, fraud detection, semantic search, and AI powered chatbot using RAG in health insurance domains.
- Designed scalable data pipelines with SQL and Python for ingestion, validation, and preprocessing, ensuring high data quality and consistency.
- Developed and optimized machine learning, deep learning, language models and advance NLP to solve business-critical problems.
- Enhanced fraud detection accuracy by integrating anomaly triggers and supervised models with a focus on minimizing false negatives.
- Delivered interpretable, production-ready solutions through Explainability techniques, APIs, and interactive dashboards for business stakeholders.

EDUCATION

Bachelor of Engineering

Ahilyadevi Holkar Solapur University

2013 - 2017 Pune, India

LANGUAGES

English
Advanced

Hindi
Advanced